

Outline

1. Introduction to ML and Linear Regression

- Introduction to AI and ML
- Features and Labels
- Supervised and Unsupervised Learning
- Regression and Classification
- ML in Chemistry
- Linear Regression

Assignment for Linear Regression

3. Core Concepts in ML and Popular Algorithms

- Overfitting
- K-fold Cross-validation
- Standardization and Regularization
- Recursive Feature Elimination (RFE)
- Probabilistic Models
- Decision Tree and Ensemble Learning

Assignment to Train a Model using ML Pipeline

2. Fingerprinting and Polynomial Regression

- Test-Train Splitting
- Feature Identification for Chemical Problems
- Experimental, Computational and Graph Features
- Introduction of Non-linearity
- Polynomial Features
- Polynomial Regression

Assignment for Polynomial Regression

4. Deep Learning and Neural Networks

- Neural Network Framework
- Activation Functions
- Forward and Backward Propagations
- Training ANN Models
- Optimization Algorithms

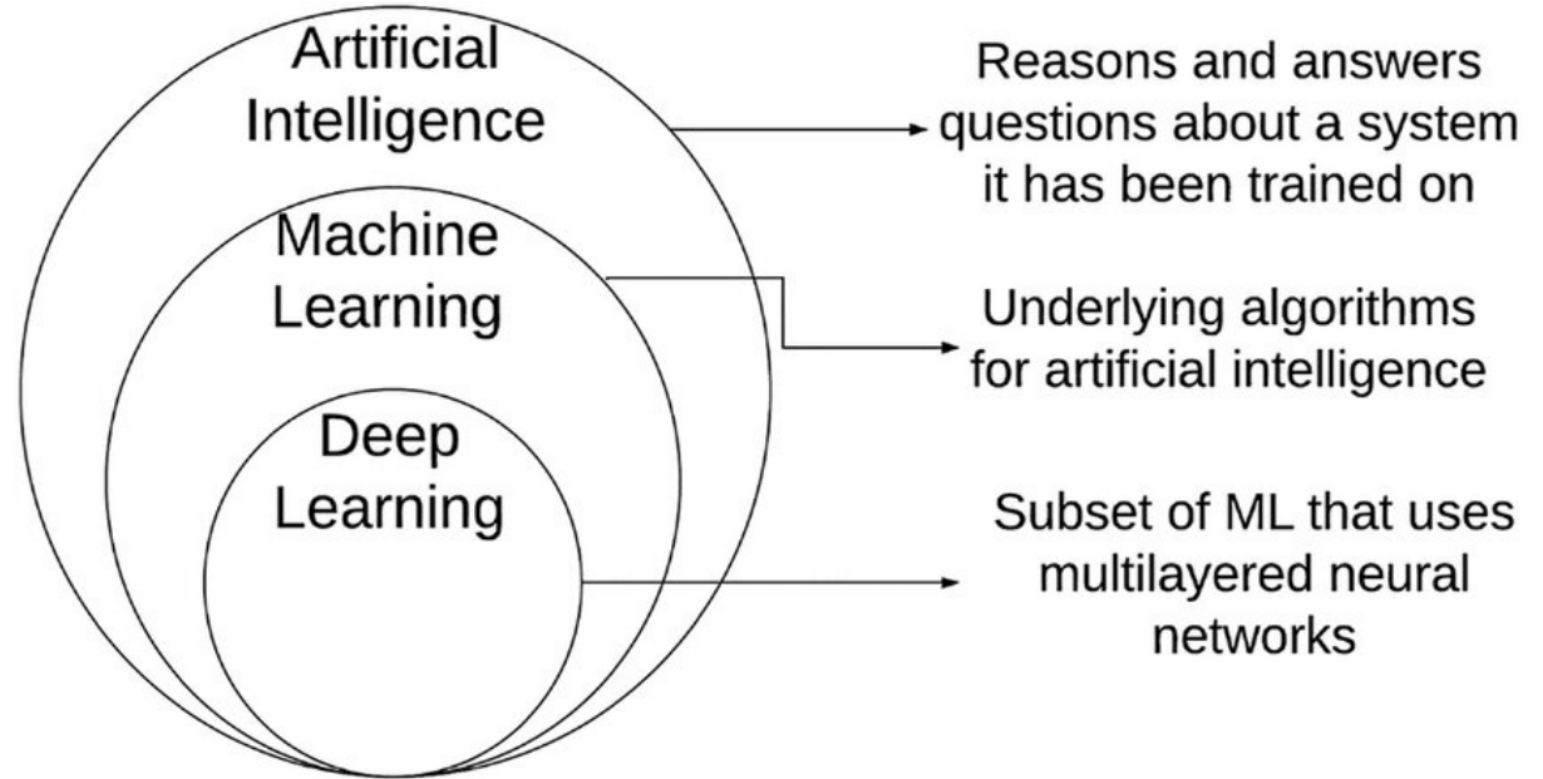
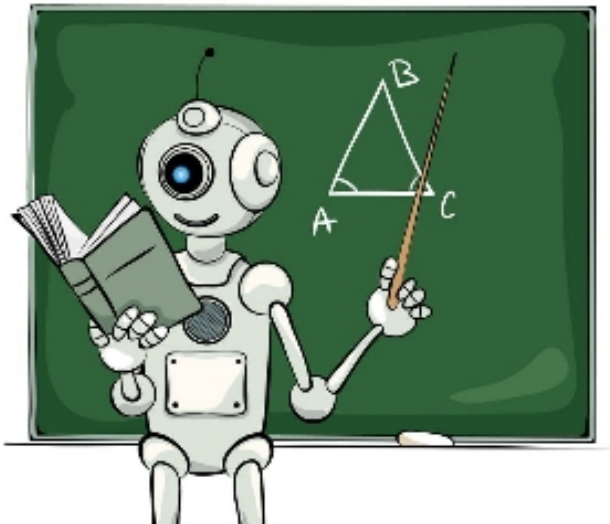
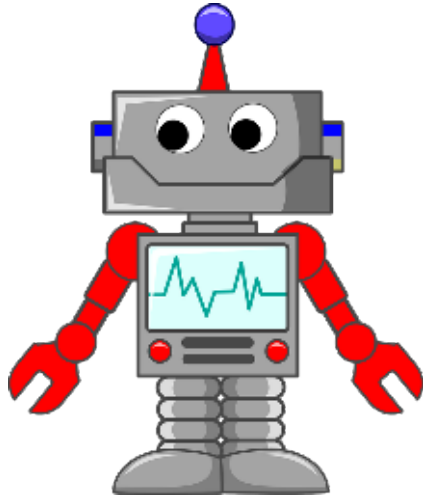
Assignment for Neural Network Model

1. Introduction to Machine Learning (ML) and Linear Regression

- Introduction to AI and ML
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Assignment for Simple Linear Regression

Artificial Intelligence and Machine Learning



- The algorithm used by AI is called a Machine Learning (ML) model
- E.g. ChatGPT uses a large language model (LLM), a neural network-based deep learning model

Features and Labels



Features	4 legs	2 legs	2 legs
	Fur	Clothes	Feathers
	1 foot tall	5 feet tall	0.5 feet tall
	Meow	Speech	Tweet
Label	Cat	Human	Bird

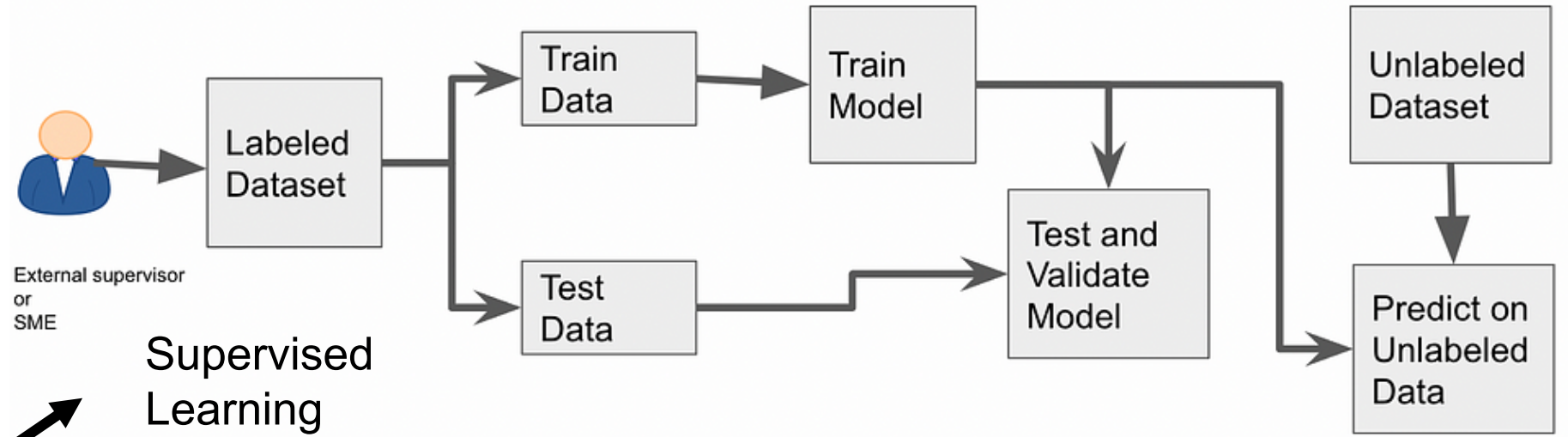
Features

- The input variables in the data we provide to the machine.
- Also known as:
 - Fingerprint
 - Descriptors
 - Independent variables
 - Predictors
 - Attributes ...etc.
- Commonly denoted by $\vec{X}$ or simply X
- E.g., Concentrations of different metals – Cu, Fe, Ni, Al, Co, Cr – in a high-entropy alloy.

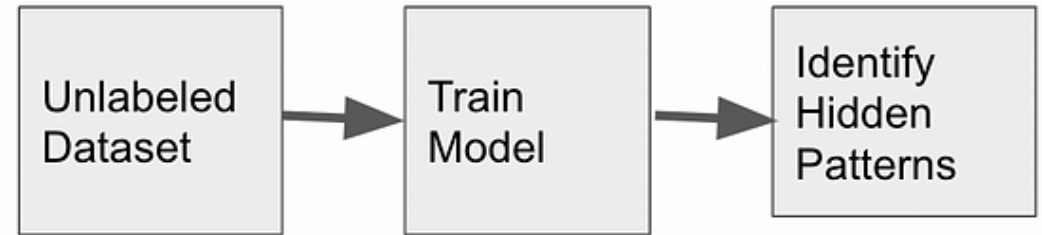
Labels

- The property of interest that we want the machine to predict.
- Also known as:
 - Target property
 - Target value
 - Dependent variable
 - Predicted variable
 - Output variable ...etc.
- Commonly denoted by $\vec{y}$ or simply y
- E.g., The hardness of the alloy.

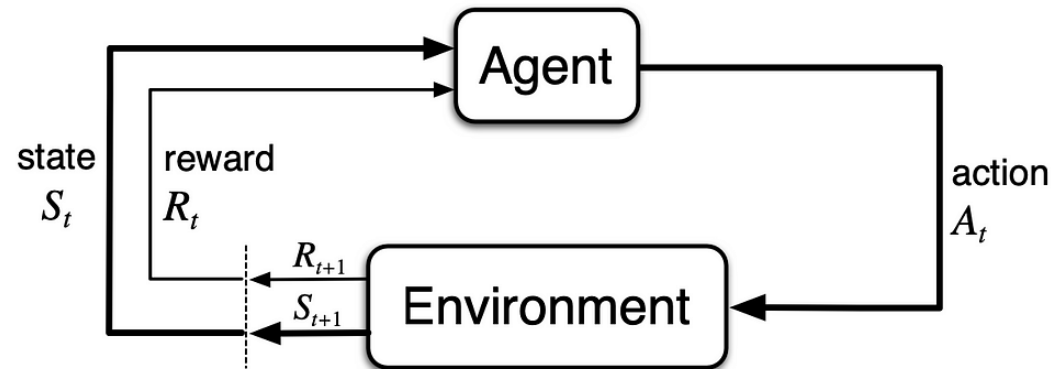
Machine Learning



Unsupervised Learning

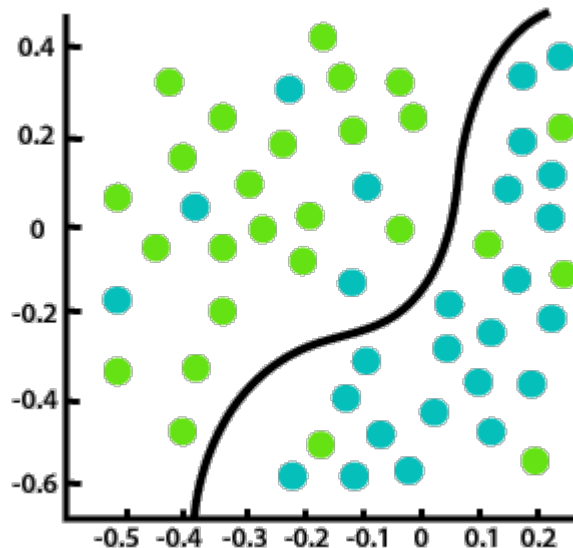


Reinforcement Learning



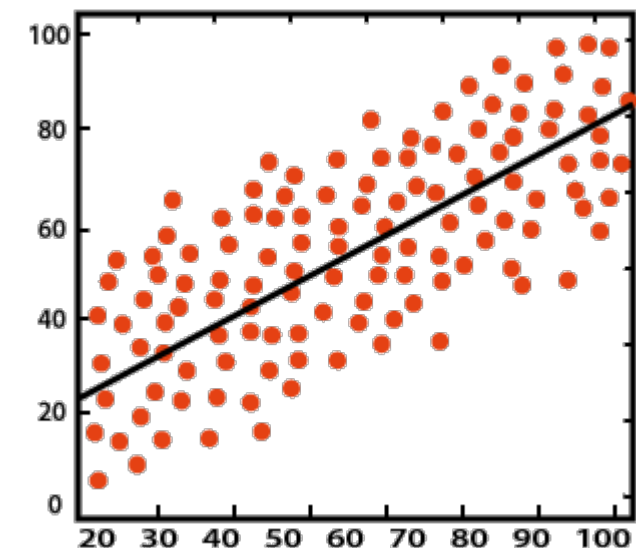
Classification

- Labels are categories or a classifications
- Labels a discrete values
- Examples:
 - Cat, human, or bird?
 - True or false? (Boolean)
 - Will it rain today or not?
 - Hard alloy or soft alloy?



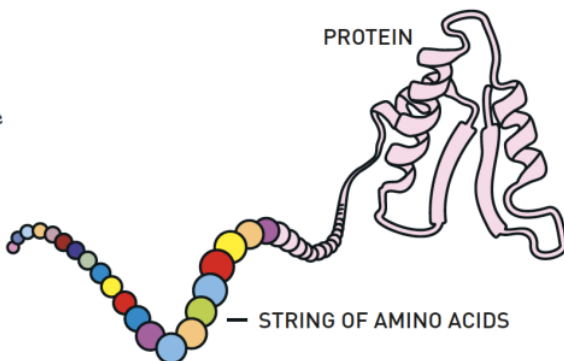
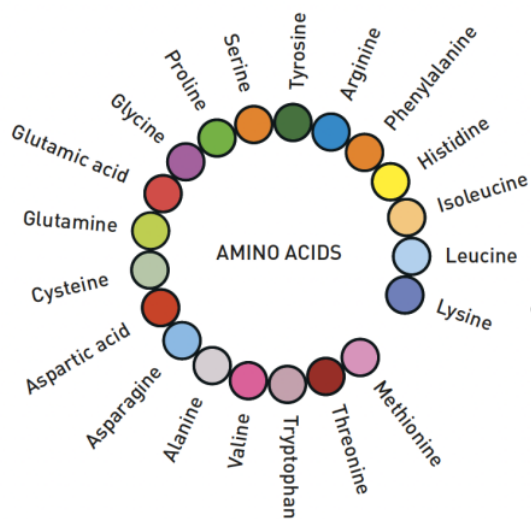
Regression

- Labels are numerical values
- Labels are usually continuous values
- Examples:
 - Number of cats in an area
 - Amount of rainfall today
 - Price of a house in Chennai
 - Hardness of an alloy



Machine Learning in Chemistry

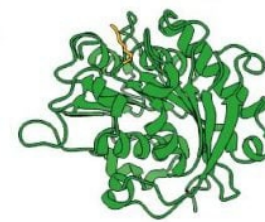
- **Example:** [AlphaFold2](#) developed by Demis Hassabis and John Jumper – a machine learning model to predict properties of proteins from the amino acid sequence.



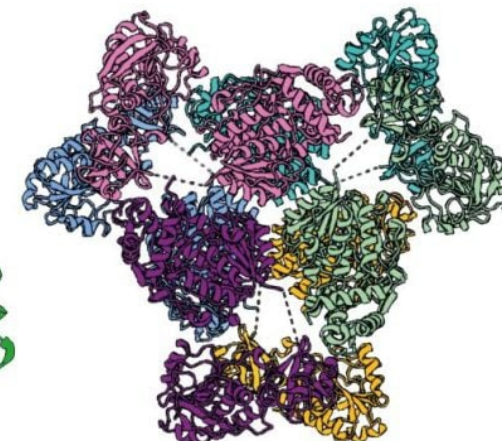
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2022: Part of a huge molecular structure in the human body. More than a thousand proteins form a pore through the membrane surrounding the cell nucleus.



2022: Natural enzymes that can decompose plastic. The aim is to design proteins that can be used to recycle plastic.



2023: A bacterial enzyme that causes antibiotic resistance. The structure is important for discovering ways of preventing antibiotic resistance.

Nobel Prize in Chemistry 2024

- **Other examples:** machine learning models to predict reaction products, mechanisms, catalytic efficiencies, optical, mechanical, electrical and thermodynamic properties.

Dataset

Data for high-entropy alloys with different concentrations of metals – Cu, Ni, Al, Co, Cr, Fe – and their hardness in HV (Vickers hardness)

Concentrations of metals

Hardness

ID	Al	Co	Cr	Cu	Fe	Ni	HV
1	0.06	0.24	0.24	0.00	0.24	0.24	110
2	0.02	0.24	0.24	0.00	0.24	0.24	118
4	0.09	0.23	0.23	0.00	0.23	0.23	127
5	0.09	0.23	0.23	0.00	0.23	0.23	131
7	0.00	0.25	0.25	0.00	0.25	0.25	139
8	0.00	0.27	0.27	0.00	0.27	0.20	140
9	0.00	0.00	0.25	0.25	0.25	0.25	143

6 Features

Al	Co	Cr	Cu	Fe	Ni
0.06	0.24	0.24	0.00	0.24	0.24
0.02	0.24	0.24	0.00	0.24	0.24
0.09	0.23	0.23	0.00	0.23	0.23
0.09	0.23	0.23	0.00	0.23	0.23
0.00	0.25	0.25	0.00	0.25	0.25
0.00	0.27	0.27	0.00	0.27	0.20
0.00	0.00	0.25	0.25	0.25	0.25

$\vec{X}$



1 Target property

HV
110
118
127
131
139
140
143

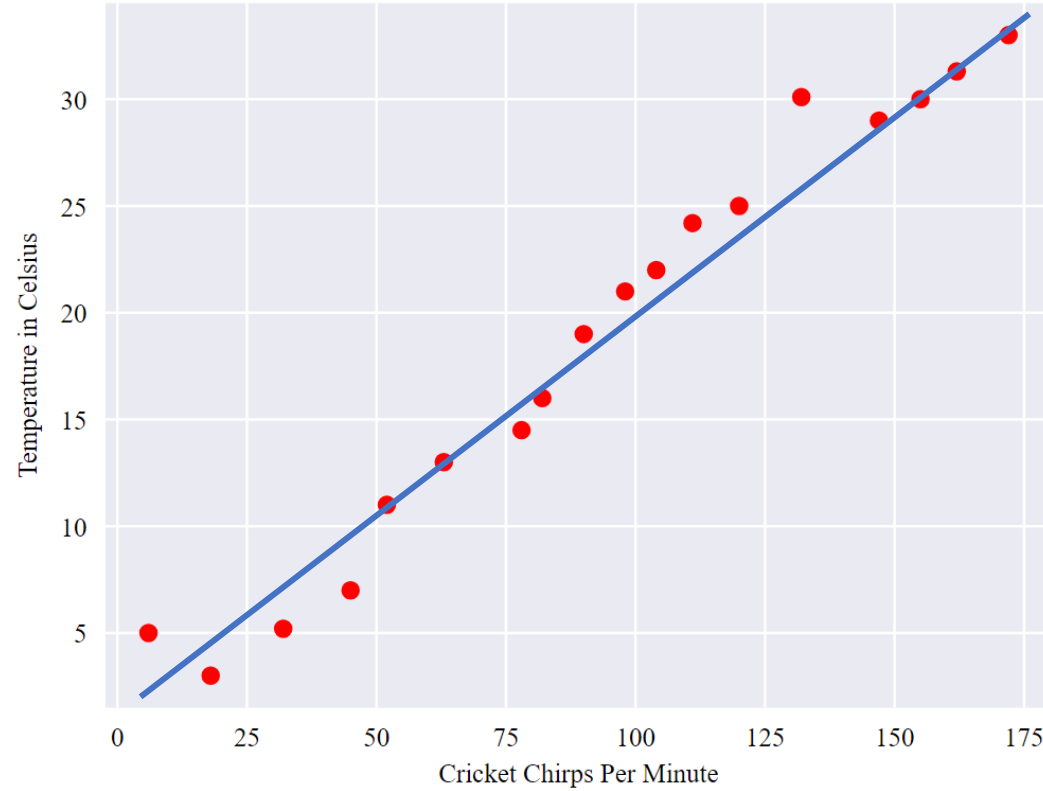
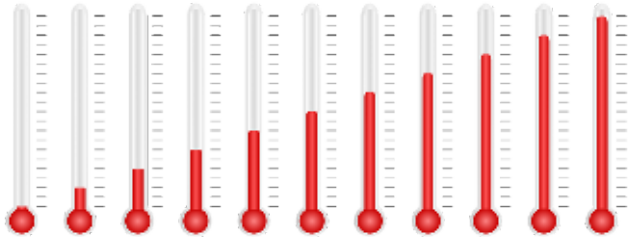
$\vec{y}$

Linear Regression



x = cricket chirps per minute

y = temperature in Celsius



Equation:

$$y = mx + c$$

Conventionally in ML,

$$y = w_0 + w_1x_1$$

For a multivariate linear regression,

$$y = w_0 + w_1x_1 + w_2x_2 + w_3x_3 + \cdots + w_Mx_M$$

y = target property

x_i = features

w_i = weights

M = number of features

Linear Regression

In the matrix form,

No. of datapoints (N) $\downarrow$

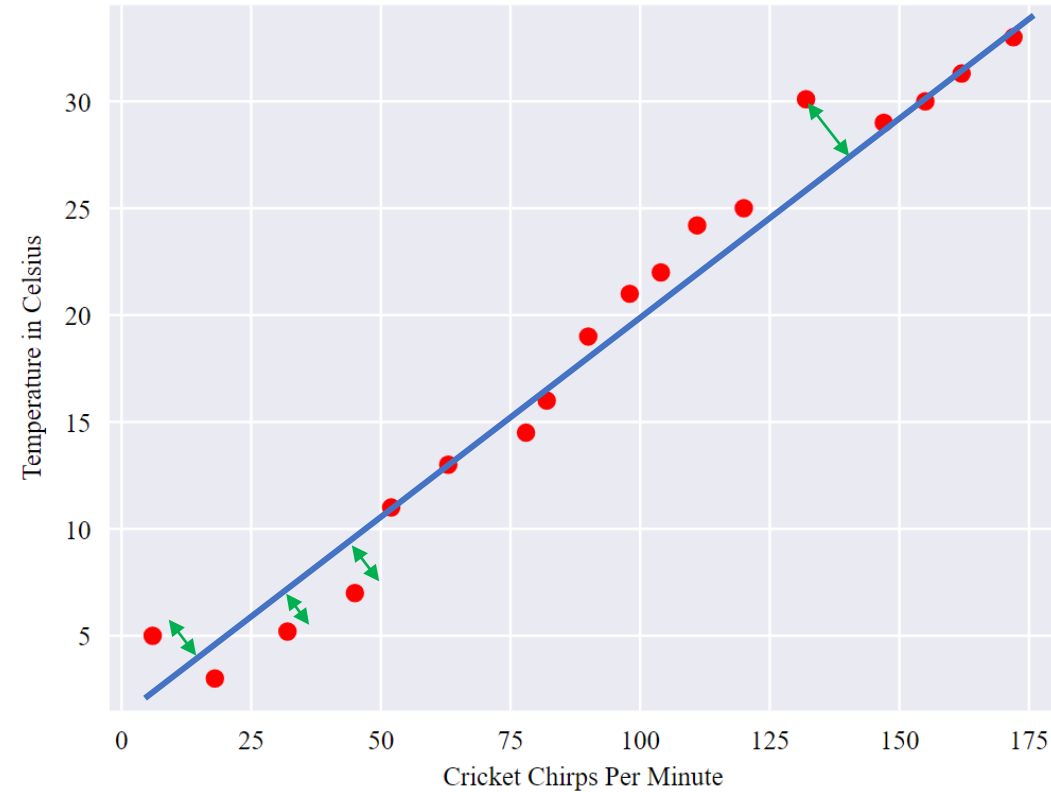
$$\begin{bmatrix} y^{(1)} \\ y^{(2)} \\ y^{(3)} \\ \vdots \\ y^{(N)} \end{bmatrix} = \begin{bmatrix} 1 & x_1^{(1)} & x_2^{(1)} & x_3^{(1)} & \dots & x_M^{(1)} \\ 1 & x_1^{(2)} & x_2^{(2)} & x_3^{(2)} & & \\ 1 & x_1^{(3)} & x_2^{(3)} & & & \\ \vdots & \vdots & & \ddots & & \\ 1 & x_1^{(N)} & & & & x_M^{(N)} \end{bmatrix}^T \begin{bmatrix} w_0 & w_1 & w_2 & \dots & w_M \end{bmatrix}$$

$\xrightarrow{\hspace{10em}}$ No. of features (M)

$$\mathbf{y} = \mathbf{X}^T \mathbf{w}$$

Weights, $\mathbf{w} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$

Loss Function or Cost Function



$$J = (\mathbf{y} - \mathbf{X}^T \mathbf{w})^T (\mathbf{y} - \mathbf{X}^T \mathbf{w}) \quad \text{(Mean Squared Error)}$$

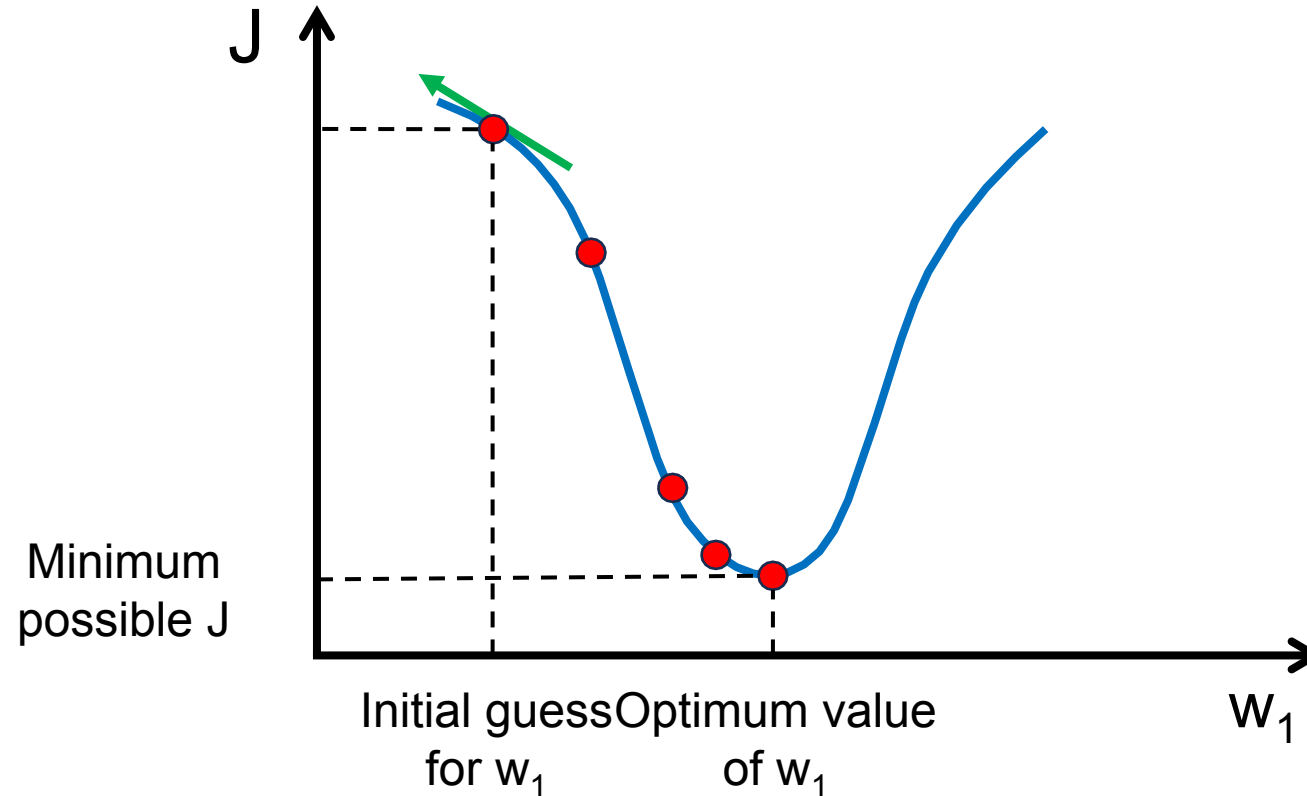
How do we get $\mathbf{w}$?

Minimize the cost function, $\nabla_{\mathbf{w}} J = 0$

$\mathbf{y}$ = True target properties
 $\mathbf{X}^T \mathbf{w}$ = Predicted target properties

Optimizing the Weights by Minimizing Cost Function

Gradient Descent



$$\nabla_w J = 0 \quad \text{or} \quad \frac{\delta J}{\delta w_1} = 0$$

α = step size

$$w_{new} = w_{old} - \alpha \frac{\delta J}{\delta w_1}$$

Useful Python Libraries

Pandas



for playing with the dataset: importing, sorting, cleaning, etc.



for all the useful matrix or array operations



for training and validating machine learning models



TensorFlow

for training and validating deep learning models such as neural networks (needs big data)

Assignment for Linear Regression

Using the provided dataset for the hardness of high-entropy alloys and their metal concentrations,

- 1) Import the datasheet (`alloy-confp-train-data.csv`) and define the data frame (`df`) using Pandas.
- 2) Separate the `df` columns into features (`X_cols`) and target property (`y_cols`).
- 3) Convert the feature columns and target property column into arrays (`X` and `y`) using NumPy.
- 4) Use `LinearRegression()` function of the `sklearn.linear_model` module to fit `X` and `y`.
- 5) Determine the RMSE and R2 score of the fitting using `root_mean_squared_error()` and `r2_score()` functions of the `sklearn.metrics` module
- 6) Determine the weights (coefficients) and the intercept using the `coef_` and `intercept_` functions.
- 7) Plot the input target values (`y`) against the target values predicted (`y_pred`) using the LR model.

THANK YOU